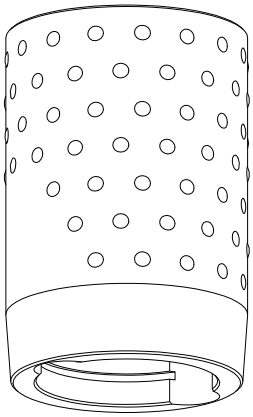


TYPE 1

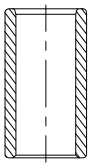


TYPE 2

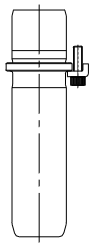
To Be Use With



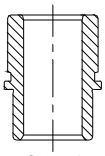
STRAIGHT
BALL BEARING
GUIDE POST



STRAIGHT SLEEVE
BALL BEARING
BUSHING

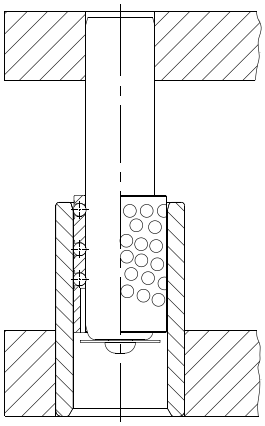


DEMOUNTABLE
BALL BEARING
GUIDE POST

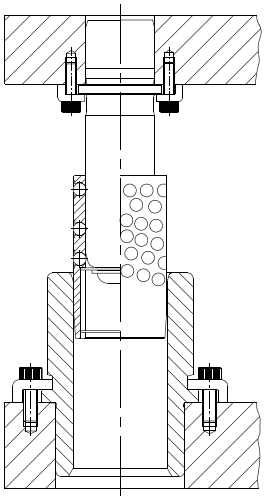


DEMOUNTABLE
BALL BEARING
BUSHING

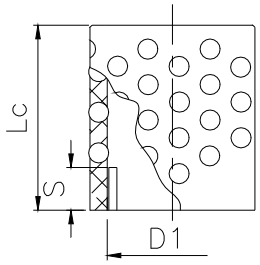
Examples



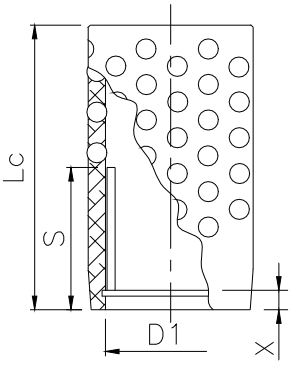
TYPE 1
PRE-LOAD



TYPE 2
PRE-LOAD RELIEVED

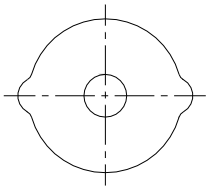


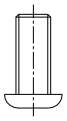
TYPE 1



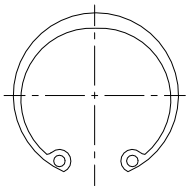
TYPE 2

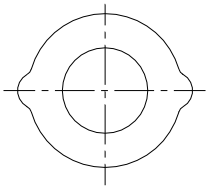
Type 1
Retainer Assembly
C01.DDD.000

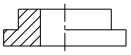


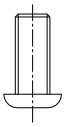


Type 2
Retainer Assembly
C02.DDD.000









MDL Ball Bearing Cages are manufactured from aluminum alloy heat treated tubing with tough and wear-resistant properties. Balls are made of bearing quality alloy steel through hardened and thoroughly inspected.

Operating Conditions:

Type 1:

In this condition at the top end of the stroke, the Guide Post does not leave the top end of the Bushing and the entire Cage remains under pre-load. At the bottom end of the stroke, the lower end of the Guide Post and the Retainer Assembly can leave the lower end of the Cage.

On type 1 condition, the Cage remains under full pre-load contact with the Guide Post and the Bushing during the entire length of the stroke.

Type 2:

In this condition at the upper end of the stroke, the Guide Post may leave the top end of the Bushing. When this happens the Cage starts to travel with the Guide Post. At the bottom end of stroke, the Guide Post and the Retainer Assembly cannot leave the lower end of the Cage due to a Snap Ring.

On type 2 condition, the Cage remains under pre-load with the Guide Post and the Bushing only when the lower end of the Guide Post goes below the top end of the Bushing.

To select Ball Bearing Cages length, see page 29 when using with Cage type 1, or page 31 when using with Cage type 2.

D_1		25	32	40	50	63	80
L_c	S	CATALOG NUMBER TYPE 1					
36	11.5	C11.025.036	C11.032.036				
48	17.5	C11.025.048	C11.032.048	C11.040.048			
60	23.5			C11.040.060			
70	28.5				C11.050.070		
84	35.5				C11.050.084		
98	42.5					C11.063.098	C11.080.098
L_c	S	CATALOG NUMBER TYPE 2					
55	27	C12.025.055					
70	35	C12.025.070	C12.032.070				
	36			C12.040.070			
85	43						
90	42	C12.025.090		C12.040.085			
	43		C12.032.090				
100	50	C12.025.100					
105			C12.032.105				
	51			C12.040.105	C12.050.105		
110	60	C12.025.110					
115			C12.032.115				
	61			C12.040.115			
120					C12.050.120		
125	70		C12.032.125				
	71			C12.040.125			
135	80		C12.032.135				
	81			C12.040.135			
140	71				C12.050.140		
145	72					C12.063.145	C12.080.145
	91			C12.040.145			
150	81				C12.050.150		
155	103			C12.040.155			
160	91				C12.050.160		
165	82					C12.063.165	C12.080.165
170	103				C12.050.170		
180	92					C12.063.180	C12.080.180
185	116				C12.050.185		
190	105					C12.063.190	C12.080.190
195	128				C12.050.195		
205	117					C12.063.205	C12.080.205
215	130					C12.063.215	C12.080.215
X		4.2		5.8	7.0		8.5

CAUTION:

Tool design with Ball Bearing Components for a certain press stroke should never be used in a press with bigger stroke.

It may be used in a press with smaller stroke.